

Assignment 10: Data Scraping

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Load the packages `tidyverse`, `rvest`, and any others you end up using.
 - Check your working directory

```
#1
library(tidyverse);library(lubridate);library(viridis);library(here);
library(rvest)
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham’s 2024 Municipal Local Water Supply Plan (LWSP):
 - Navigate to <https://www.ncwater.org/WUDC/app/LWSP/search.php>
 - Scroll down and select the LWSP link next to Durham Municipality.
 - Note the web address: <https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024>

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

#2

```
webpage <- read_html('https://www.ncwater.org/WUDC/app/LWSP/report.php?pswid=03-32-010&year=2024')
```

3. The data we want to collect are listed below:

- From the “1. System Information” section:
- Water system name
- PSWID
- Ownership
- From the “3. Water Supply Sources” section:
- Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

HINT: The first value should be “Durham”, the second “03-32-010”, the third “Municipality”, and the last should be a vector of 12 numeric values (represented as strings)“.

#3

```
name <- webpage %>% html_nodes('div+ table tr:nth-child(1) td:nth-child(2)') %>%  
  html_text()  
PSWID <- webpage %>% html_nodes('td tr:nth-child(1) td:nth-child(5)') %>%  
  html_text()  
ownership <- webpage %>% html_nodes('div+ table tr:nth-child(2) td:nth-child(4)') %>%  
  html_text()  
MGD <- webpage %>% html_nodes('th~ td+ td') %>%  
  html_text()
```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

TIP: Use `rep()` to repeat a value when creating a dataframe.

NOTE: It’s likely you won’t be able to scrape the monthly withdrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: “Jan”, “May”, “Sept”, “Feb”, etc...

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

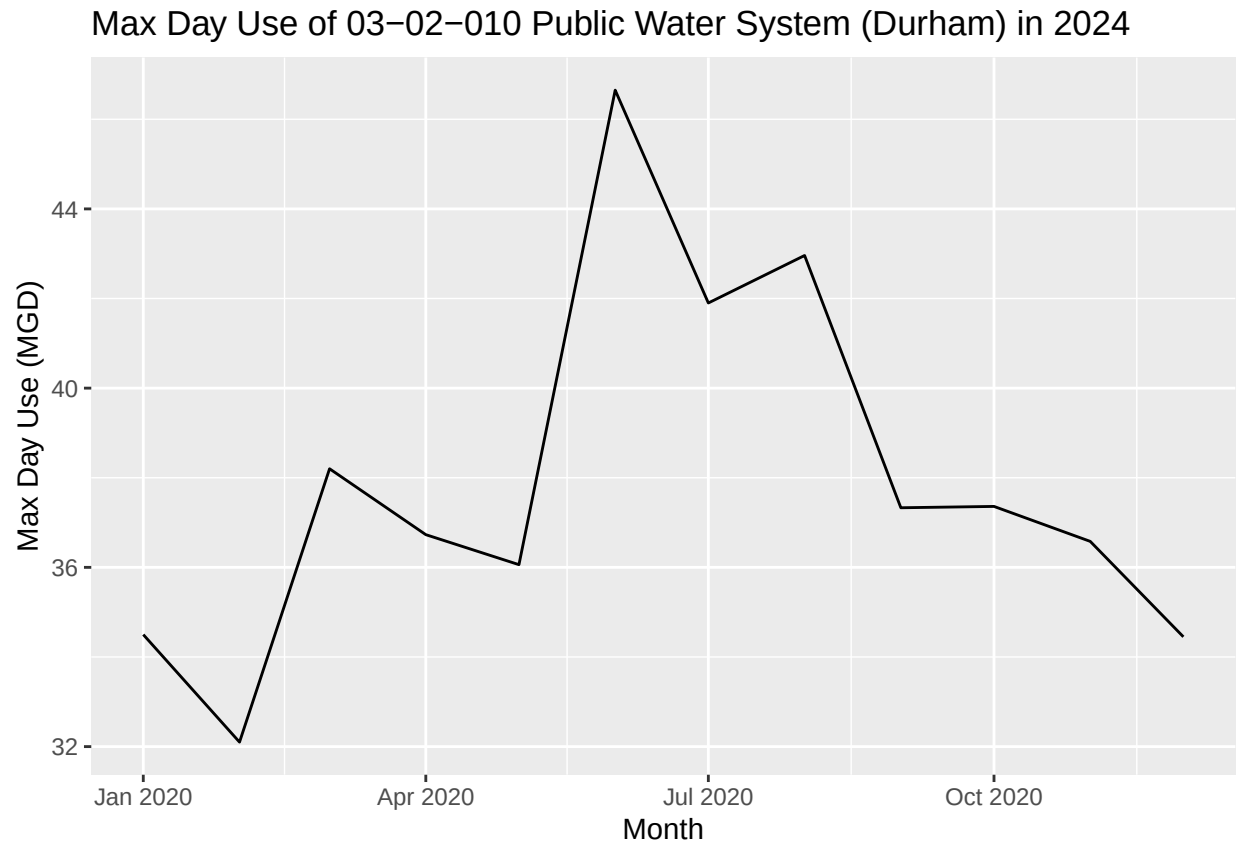
#4

```
df_watersupply <- data.frame("Month" = c(1,5,9,2,6,10,3,7,11,4,8,12),  
                             "Water system name" = rep(name),  
                             "PSWID" = rep(PSWID),  
                             "Ownership" = rep(ownership),  
                             "MGD" = as.numeric(MGD)) %>%
```

```
arrange(Month) %>%
mutate(Date=mdy(paste(Month, 1,2020,sep="-")))
```

#5

```
ggplot(df_watersupply,aes(x=Date,y=MGD)) +
  geom_line()+
  labs(x = "Month",y = "Max Day Use (MGD)")+
  ggtitle("Max Day Use of 03-02-010 Public Water System (Durham) in 2024")
```



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs - “PWSID” and “year” - that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function’s output

#6.

```
scraped.dataframe<- function(PWSID,year){
  the_website<-read_html(paste0(
    'https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=',
```

```

PWSID, '&year=', year))

name <- the_website %>% html_nodes(
  'div+ table tr:nth-child(1) td:nth-child(2)') %>%
  html_text()
PSWID <- the_website %>% html_nodes(
  'td tr:nth-child(1) td:nth-child(5)') %>%
  html_text()
ownership <- the_website %>% html_nodes(
  'div+ table tr:nth-child(2) td:nth-child(4)') %>%
  html_text()
MGD <- the_website %>% html_nodes(
  'th~ td+ td') %>%
  html_text()

df_scraped <- data.frame("Month" = c(1,5,9,2,6,10,3,7,11,4,8,12),
                        "Year" = rep(year),
                        "Water system name" = rep(name),
                        "PSWID" = rep(PSWID),
                        "Ownership" = rep(ownership),
                        "MGD" = as.numeric(MGD)) %>%

arrange(Month) %>%
  mutate(Date=mdy(paste(Month, 1,year,sep="-")))
return(df_scraped)
}

```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```

#7
Durham.2020<-scraped.dataframe('03-32-010',2020)
view(Durham.2020)

```

8. Use the function above to extract data for Cary (PWSID = '03-92-020') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

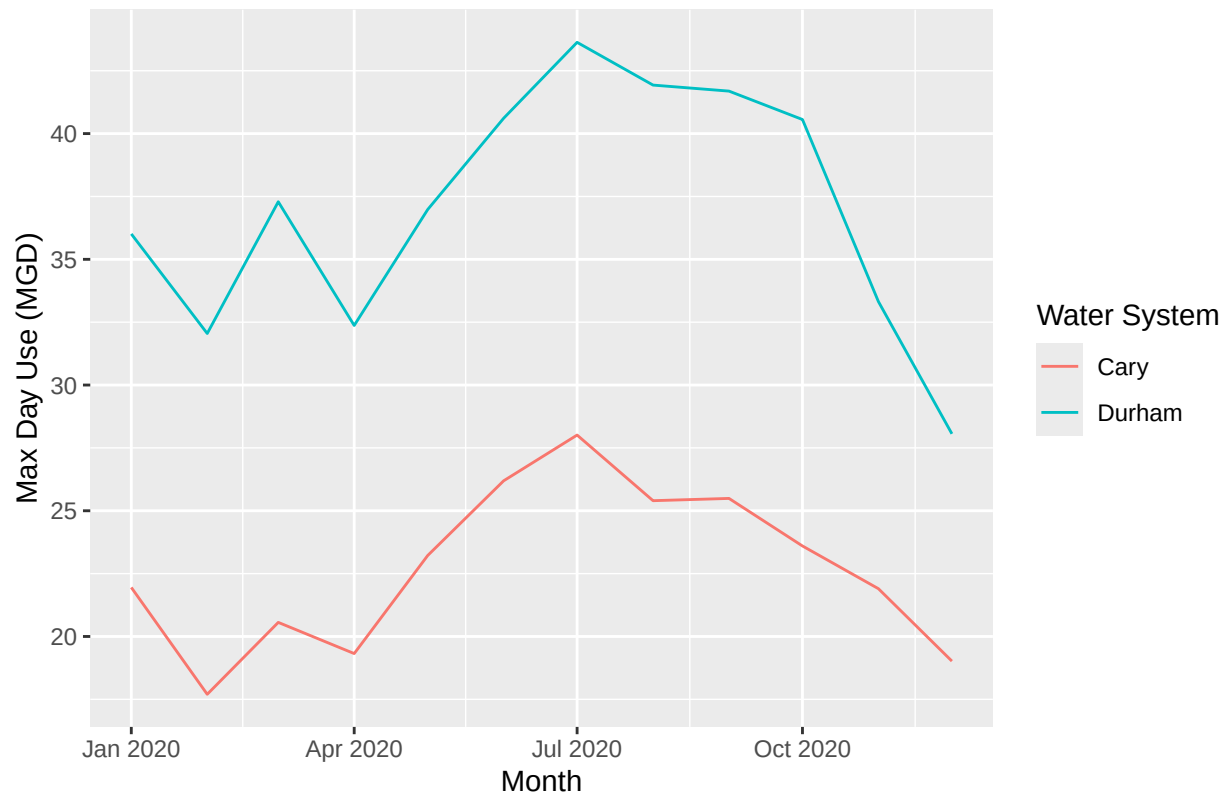
```

#8
cary.2020<-scraped.dataframe('03-92-020',2020)
DurhamCary.2020<-bind_rows(cary.2020, Durham.2020)

ggplot(DurhamCary.2020,aes(x=Date,y=MGD,
                          color=Water.system.name)) +
  geom_line()+
  labs(x = "Month",y = "Max Day Use (MGD)", color='Water System')+
  ggtitle("Max Day Use of Durham and Cary Public Water Systems in 2020")

```

Max Day Use of Durham and Cary Public Water Systems in 2020



- Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = 'loess').

TIP: See Section 3.2 in the "10_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs. Pipe the output of the map2() function to bind_rows() to combine the dataframes into a single one, and use that to construct your plot.

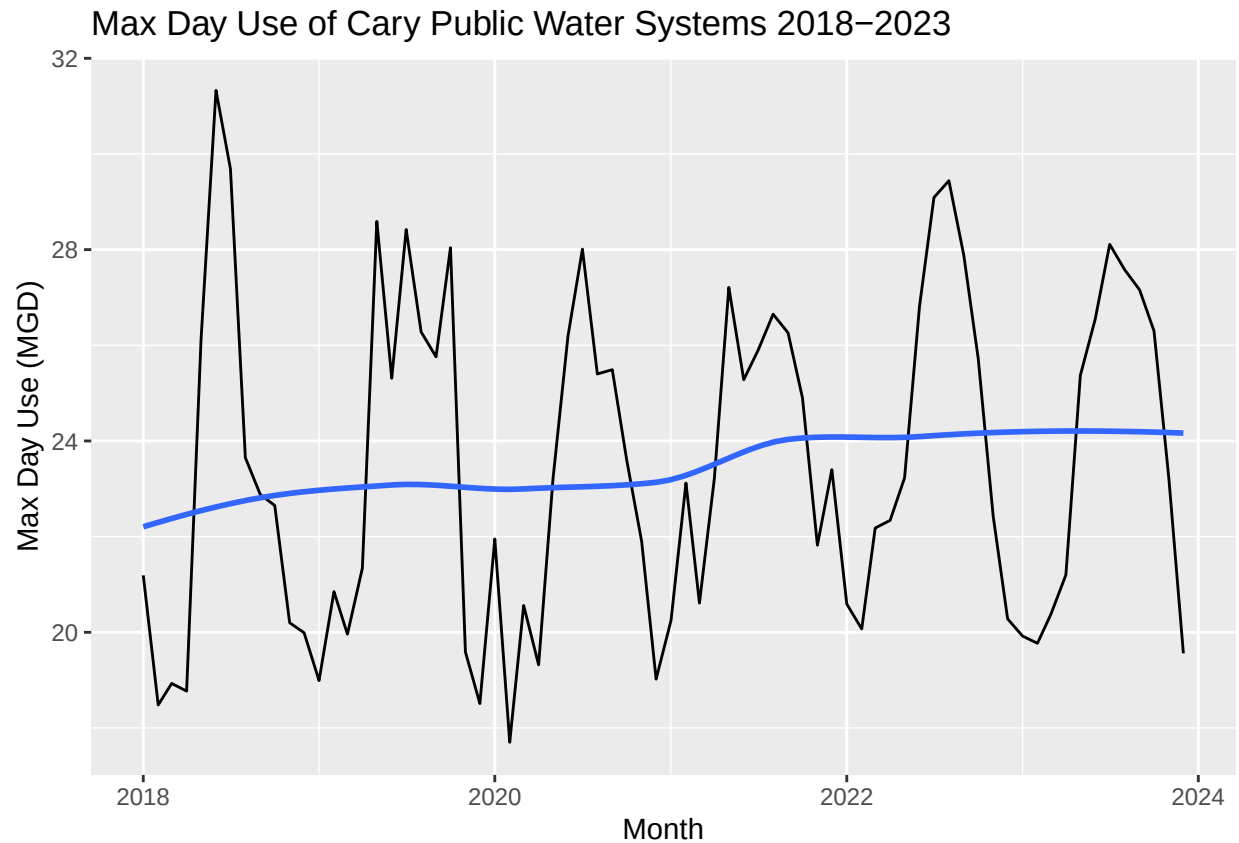
```
#9
the_years <- c(2018,2019,2020,2021,2022,2023)

#"Map" the "scrape.it" function to retrieve data for all these
dfs_Cary <- map2('03-92-020', the_years, scraped.dataframe)

#Conflate the returned list of dataframes into a single one
df_Cary <- bind_rows(dfs_Cary)

#plot over the years
ggplot(df_Cary, aes(x=Date, y=MGD)) +
  geom_line() +
  geom_smooth(method = "loess", se = FALSE) +
  labs(x = "Month", y = "Max Day Use (MGD)") +
  ggtitle("Max Day Use of Cary Public Water Systems 2018-2023")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: >Yes, there seems to be a seasonal trend in water use in Cary. Seems to peak in the summer (Months 5-9) and decrease for the winter. There seems to be a slight increasing overall trend from 2018-2023 as shown by the smoothed line.